ANMOL RATTAN SINGH SANDHU

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EDUCATION

Olin College of Engineering | B.S. in Engineering (Computing), GPA: 3.9/4.0 — May 2025

Relevant Coursework: Advanced Algorithms, Software Systems, Computational Robotics, Neurotechnology and ML, Data Science, Collaborative Design, Computer Architecture, Longer Term Software Development

SKILLS

- Python, Bash, SQL, Go, JavaScript, Java, C, C++, Rust, Kotlin, Dart, R, MATLAB
- Git, GitHub, Linux, Docker, AWS, Pydantic, SQLAlchemy, Kubernetes, React.js, React Native, Ansible, Firebase, AZ-900

PROFESSIONAL EXPERIENCE

Indico Data Solutions

- **Software Engineer II** — Aug 2025 – Present
  - Built an extensible notification framework with **Amazon SQS** as its first delivery backend, supporting **IAM** role assumption for same- and cross-account deployments. Moved notification configuration from static, cluster-level YAML files to schema-validated, API-managed settings and added database-backed delivery tracking.
  - Built a custom heartbeat system for the **async task queue** powering Indico's **agentic decisioning platform**, allowing stuck workers to be detected and recovered automatically in minutes instead of hours.
  - Designed and implemented a **FastAPI REST** gateway over an existing **GraphQL** service, unblocking a third-party integration by providing **OAuth**-secured, **Pydantic**-validated REST endpoints.
- **Software Engineer Intern** — May 2025 – Aug 2025
  - Contributed bug fixes and reliability improvements across backend microservices.

Research Assistant, MIT CSAIL (FutureTech Lab) — Jun 2024 – Dec 2025

- Researched scaling laws for **quantization in LLMs**, focusing on the tradeoffs between efficiency and accuracy that emerge under different model-compression methods.
- Contributed to the **Algorithm Wiki project**, an open resource documenting algorithmic development and complexity trends.

Technical Lead, Senior Capstone Program in Engineering (SCOPE), Olin College — Sept 2024 – May 2025

- Collaborated with Boston University and Red Hat to prototype **LLM-powered agents** for personalized reading-comprehension tools for K–3 students.
- Designed the foundational infrastructure of the application and implemented the core pipeline integrating **Llama 3.1** on **TorchServe**, a **React/Next.js** front end, and **PostgreSQL** for user data and conversation context.

Intern, Modular Open-Source Identification Platform (MOSIP) — Jan 2024 – May 2024

- Improved the open-source **Bluetooth** credential exchange module, Tuvali, of INJI, allowing a presenter to select from a list of verifiers and improving the previous QR-code scanning process used to connect to verifiers.
- INJI, MOSIP's decentralized credential wallet, enables users to manage and verify **OpenID**-conforming credentials. MOSIP is an open-source, Aadhaar-inspired identity platform that has helped issue more than 100M digital IDs worldwide.

Full Stack Developer (Volunteer), Community Knights (Non-Profit) — Jun 2023 – Dec 2023

- Developed an accessible ride-sharing platform for vulnerable populations with Community Knights.
- Utilized **React.js**, **React Native**, and **Firebase** to create applications with **CRUD** operations and **RBAC**.
- Conducted UX design interviews to iteratively improve the application.

PROJECTS

- **Rust-EDIS:** A scalable, distributed key-value store built in **Rust**, implementing a reader-writer shard model.
- **Clipboard-Transformer:** A simple tool built in **C++** to transform text stored in the clipboard.
- **Image Segmentation:** Separates images into distinct segments using graph-cut algorithms and network flows. Built in **Python** with **NetworkX** and **OpenCV**.
- **Huffman Encoding:** A compression algorithm implemented in **C++**.
- **CNN-MNIST:** Classifies handwritten digits from the MNIST dataset using only **NumPy** and **Python**.
- **Sudoku Solver:** Solves Sudoku puzzles using the **simulated annealing** algorithm. Implemented in **Python**.

EMPLOYMENT, LEADERSHIP AND INTERESTS

- Instructor, Advanced Algorithms (Student-Led Course)
- Resident Advisor, Olin College of Engineering
- Sub-team Lead, Public Interest Technologies Club, Olin College of Engineering
- Vice President, Olin South Asian Student Organization
- Interests: Rock climbing, badminton, and swimming